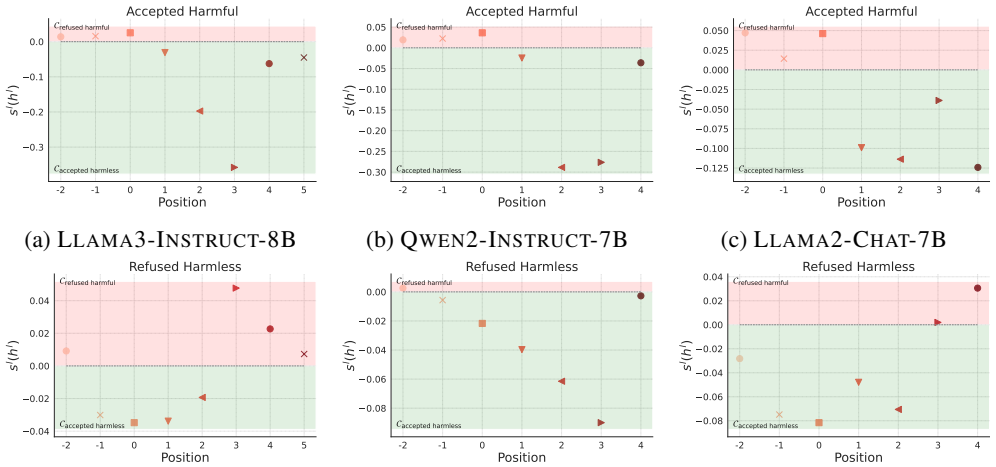
Figure 13: Different token positions where we extract hidden states for comparison experiments on Llama2. We consider extra two tokens before t_{inst} and all the tokens till $t_{\text{post-inst}}$.



(a) LLAMA3-INSTRUCT-8B (b) QWEN2-INSTRUCT-7B (c) LLAMA2-CHAT-7B

(d) LLAMA3-INSTRUCT-8B (e) QWEN2-INSTRUCT-7B (f) LLAMA2-CHAT-7B

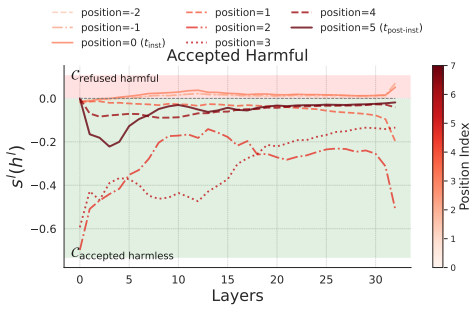
Figure 14: Average $s^l(h^l)$ over the middle layers for hidden states extracted at different token positions. $s^l(h^l)$ indicates the difference between the cosine-similarity with the cluster of refused harmful instructions and the cluster of accepted harmless instructions. Larger $s^l(h^l)$ indicates the hidden state is closer to $\mathcal{C}_{\text{refused harmful}}$. If the token position encodes harmfulness correctly rather than refusal features, the hidden states of accepted harmful instructions should fall in the red region (high $s^l(h^l)$), while those of refused harmless ones should fall in the green region (small $s^l(h^l)$). Only position t_{inst} satisfies both requirements at the same time for all models.

F Analysis on More Token Positions

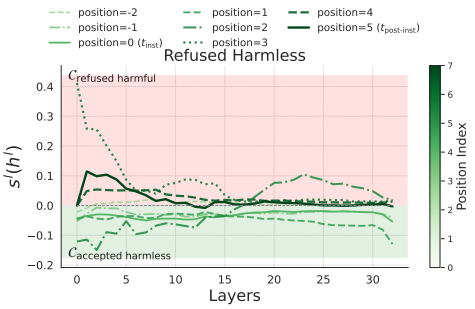
In this section, we investigate LLMs' hidden states extracted from token positions other than t_{inst} and $t_{\text{post-inst}}$. We extract tokens starting from the position right before t_{inst} until $t_{\text{post-inst}}$ as illustrated in Figure 13. We first compare the clustering patterns of hidden states extracted at each token position. We then study the steering performance with directions extracted at each token position. Our results suggest that t_{inst} position encodes harmfulness the most evidently.

F.1 Clustering at different token positions

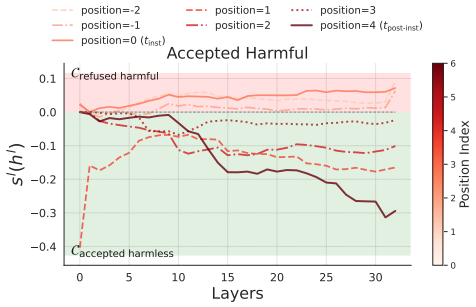
Following implementations in Section 3.2, at each token position, we compute $s^l(h^l)$ by Equation 2 to measure whether the hidden states of test cases h^l at layer l fall in the cluster of rejected harmful instructions or accepted harmless instructions. Larger positive $s^l(h^l)$ means these hidden states are closer to the cluster of refused harmful instructions. In Figure 14, we average the layer-wise $s^l(h^l)$ in the middle layers (9 to 20) as they tend to be more responsible for handling harmfulness information inside LLMs (evidenced by observation that the steering performance reaches peak in the middle layers in experiments of Section 3.4 and Section 3.5). If a token position encodes harmfulness, then the clustering of examples in the latent space should reflect the shared harm-related features instead of the refusal-related features. Specifically, the hidden states of accepted harmful instructions should fall within the red region (the cluster of refused harmful instructions), while the refused harmless instructions should fall within the green region (the cluster of accepted harmless instructions). Among all the token positions tested, only t_{inst} demonstrates this desired clustering pattern. Full layer-wise results are shown in Figure 15.



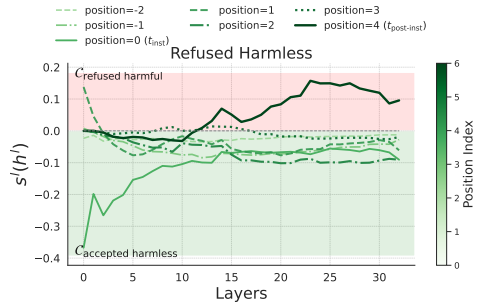
(a) LLAMA3-INSTRUCT-8B



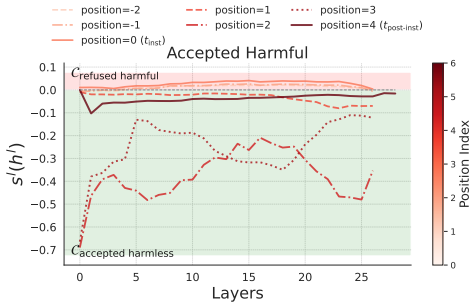
(b) LLAMA3-INSTRUCT-8B



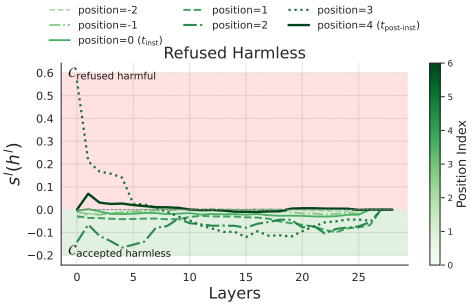
(c) LLAMA2-CHAT-7B



(d) LLAMA2-CHAT-7B

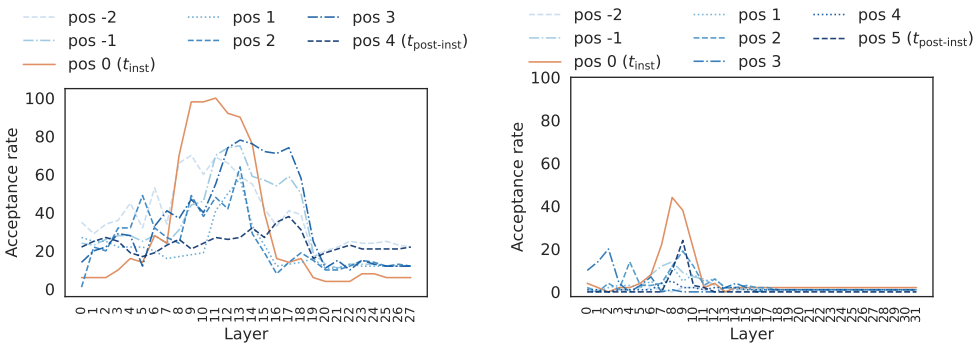


(e) QWEN2-INSTRUCT-7B



(f) QWEN2-INSTRUCT-7B

Figure 15: Clustering at different token positions



(a) Qwen2

(b) Llama3

Figure 16: Steering harmless instruction in the reply inversion task with directions between harmful instructions and harmless instructions extracted at different token positions. If the direction encodes harmfulness, our intervention should trigger the model to flip 'No' to 'Certainly' as the model may perceive the harmless instruction as harmful, leading to increased acceptance rate.

F.2 Directions extracted at different token positions

We provide further evidence by extracting steering directions at different token positions for the reply inversion task. Specifically, we extract directions from the cluster of harmless instructions to the cluster of harmful instructions at each token position following Section 3.4. We then apply those directions to the tokens before the inversion question to assess how strongly each direction raises the LLM's perception of harmfulness. Results are shown in Figure 16. For both Qwen2 and Llama3, the steering direction extracted at t_{inst} (i.e., position 0) achieves the strongest intervention effect: the model is more likely to interpret the originally harmless instructions as harmful, thereby triggering an acceptance response in the reply inversion task.